WINGS

Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines a distinctive biomimetic “wings” form designed to provide daylight and views for a dense residential programme; mineral-wool insulation, efficient common-area lighting and electric-vehicle charging; and underground waste-sorting infrastructure and approximately 3,500 m² of green space. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Strada Frunzișului 88, Cluj-Napoca, Romania
Building use / function	High-rise multi-family residential development
Project type (new build, refurbishment, extension)	New build
Plot area / Floor area	Approximately 14,820 m²; 228 apartments
Year / phase	Pre-certified in 2019; final certification / completion reported in 2021
Climate zone	Humid continental
Thematic focus / innovation	Biomimetic architecture; mineral-wool envelope; daylight; EV charging; underground waste sorting; extensive green space
Investor / building owner	Studium Green
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; façade, mobility, waste and landscape specialists
Contact person	Studium Green - https://studiumgreen.ro/
Further information	https://green-homes.org/property/wings/ https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Wings is a new build project in Strada Frunzișului 88, Cluj-Napoca, Romania. It functions as high-rise multi-family residential development. Its main best-practice contribution is Uses expressive high-rise architecture to make a comprehensive Green Homes strategy visible at city scale. The documented strategy brings together a distinctive biomimetic “wings” form designed to provide daylight and views for a dense residential programme, mineral-wool insulation, efficient common-area lighting and electric-vehicle charging, and underground waste-sorting infrastructure and approximately 3,500 m² of green space. RoGBC sources consistently identify 228 apartments; public technical detail is qualitative rather than a measured operational dataset. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - A distinctive biomimetic “wings” form designed to provide daylight and views for a dense residential programme.
2. Integrated systems and operation - Mineral-wool insulation, efficient common-area lighting and electric-vehicle charging.
3. Wider value - Underground waste-sorting infrastructure and approximately 3,500 m² of green space.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.